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The Digital Divide, Wealth, and Inequality: An Examination of Socio-Economic Determinants of Collaborative Environmental Governance in Thailand through Provincial-Level Panel Data Analysis

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Abstract: Collaborative environmental governance (CEG) is a tripartite process that engages the government, private sector, and general public in decision-making related to environmental challenges, focusing on fostering more sustainable and efficient solutions. Understanding the specific factors influencing the degree of CEG presents a significant challenge, particularly in developing countries. This study aims to identify and assess the socio-economic determinants affecting the degree of CEG in Thailand, a representative developing country. Utilizing robust panel data models, which are well-suited to handle the complex variability of socio-economic factors, we analyzed provincial-level data from 2017 and 2019. Our findings revealed the associations between the degree of CEG and variables such as the internet access divide, economic activities, income inequality, and budget allocations for environmental activities. This research fills critical gaps in our understanding of how these determinants shape collaborative governance efforts, offering novel insights that challenge existing paradigms and providing actionable recommendations for policymakers striving to enhance environmental governance in developing regions.

Keywords: collaborative environmental governance; environmental management; collaborative governance; Thailand; panel data; digital divide



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1. Introduction

The growing awareness of climate change and the pressing need to address environmental challenges have caused sustainable development to become many countries' main concern in recent years. A promising strategy to tackle these complex, multi-faceted issues involves the use of Collaborative Environmental Governance (CEG). This form of governance aims to pool resources and insights from multiple stakeholders—including governments, corporations, and the general public—in order to enact more effective and democratically accountable environmental policies [1].

CEG has been recognized as a valuable approach in environmental management due to its capacity to offer a wide array of adaptable response strategies that are capable of adjusting to dynamic conditions [2]. By fostering more legitimate and informed processes of collective decision-making and more efficient ways of managing compliance, CEG can contribute to more effective environmental policy [3]. Indeed, research has demonstrated

that governance regimes that prioritize collaboration between environmental and industry stakeholders achieve superior environmental outcomes [4]. Despite CEG's significant potential, the complexities it entails may lead to conflicts and stalemates between various stakeholders involved in environmental decision-making processes, emphasizing the need for effective frameworks and further research to guide its successful implementation [5]. In developing countries like Thailand that are grappling with severe environmental threats, there is a particularly pressing need for studies examining the application and effectiveness of CEG.

Thailand presents an interesting case study for the examination of CEG, as the country is currently contending with a variety of substantial environmental challenges. These range from air and water pollution, which costs approximately 1.6–2.6% of the nation's Gross Domestic Product (GDP) annually [6], to problems linked to rapid economic development, such as declining wildlife populations, deforestation, soil erosion, and water scarcity. In addition, a study by Attavanich [7] determined that the social cost of PM_{2.5} (Particulate Matter with a diameter of 2.5 µm or smaller) pollution across Thailand in 2019 amounted to 2.17 trillion THB (Thai baht) annually. This equates to nearly 11% of the nation's GDP for the same year. Climate change, manifesting as increased temperatures and associated threats like extreme heat and rising sea levels, poses another major challenge. Despite these pressing issues, Thailand has shown commitment to environmental protection and sustainability. The country has engaged in collaborative environmental efforts, such as partnerships with the International Union for Conservation of Nature (IUCN). These collaborative efforts aim to address environmental degradation, explore sustainable solutions, and meet international sustainability targets. Thailand has committed to ambitious goals, such as increasing its forest cover to 40% within 20 years [8] and signing the 2015 Paris Agreement, in which it pledged to reduce its annual greenhouse gas emissions by 20–25% by 2030. However, despite these commitments, Thailand's performance on various environmental parameters, as highlighted by the 2022 Environmental Performance Index (EPI), indicates a need for more effective strategies and measures. Thailand's EPI rank, which evaluates the efforts of countries to safeguard environmental well-being, boost ecosystem resilience, and counteract climate change impacts, is 108th out of 180 countries, showing room for considerable improvement. Thailand particularly demonstrates low scores in areas such as climate change mitigation, air quality, waste management, biodiversity and habitat, and ecosystem services. For instance, Thailand is among the top five producers of ocean plastic pollution with Indonesia, India, the USA, and Brazil. Collectively, these countries are responsible for 43% of the total global ocean plastic pollution [9]. Countries in Southern Asia and Asia-Pacific, including Thailand, Vietnam, and the Philippines, have also seen significant habitat loss due to infrastructure development and deforestation. This, combined with the environmental impact of tourism, has negatively impacted their biodiversity and habitat scores [10,11].

In this context, this study aims to investigate the factors influencing the degree of CEG in Thailand. Understanding the CEG degree and its influencing factors directly affects the effectiveness of environmental management strategies and policies. The degree will involve multiple stakeholders and ensure diverse perspectives. The degree of CEG can enhance the comprehensiveness and adaptability of responses to environmental issues. In the context of Thailand, a greater CEG could address pressing environmental threats more efficiently by leveraging the resources and capabilities of the government, the private sector, and the general public. Ultimately, a robust CEG can booster Thailand's pursuit of sustainable development, addressing environmental degradation while also considering socio-economic development needs. Hence, this study will contribute to the body of knowledge on environmental governance by providing a comprehensive analysis using panel data analysis with a robustness test. It will offer insights that are not only relevant to Thailand but also transferable to regions possessing similar socio-economic and environmental contexts. Furthermore, this study will serve as a bridge between quantitative

research and policymaking by providing practical recommendations for enhancing CEG, especially in the developing country context.

In the face of these issues, this study aims to assess the degree of CEG across provinces in Thailand and analyze the socio-economic factors that may influence it. As such, we formulate the following main research question: “What are the factors affecting the degree of CEG in Thailand”? To address this research question, this study uses panel data from the years 2017 and 2019 and employs the panel data approach to estimate the relationships between these factors and the degree of CEG. Addressing this question will illuminate the dynamics of CEG in Thailand, contributing to a deeper understanding of its potential role in advancing environmental sustainability. By identifying the variables that influence the effectiveness of CEG, our study will offer practical insights to guide the formation of policies and initiatives in Thailand and beyond.

This study is organized as follows. The next section presents a literature review and is followed by the formulation of hypotheses in Section 3. The research methodology, including data sources and model specifications, is described in Section 4. Section 5 provides the results from the empirical analysis. Section 6 presents the results of the robustness test. The final section, Section 7, presents a discussion and our conclusions.

2. Literature Review

2.1. Defining the CEG

Collaborative Environmental Governance (CEG) is conceptualized as a multi-actor system for addressing environmental challenges that invites participation from the government, the private sector and the general public [1]. The approach emphasizes the integration and synergistic use of the strengths and resources of each actor to achieve superior environmental outcomes. In other words, this collaborative enterprise embraces a rich diversity of stakeholders, not only government entities but also active participation from the general public, the private and non-governmental sectors, who aim to achieve commonly agreed environmental goals collectively [12]. Through collaboration, these stakeholders can often achieve more together than they could manage individually, highlighting the power of collective action in environmental governance.

The nature of CEG encourages collaboration across multiple jurisdictional levels, which fosters shared responsibility for management decisions, promotes trust, and enables social learning [13]. Such collaborations often lead to the creation of adaptive, innovative, and mutually agreed solutions to environmental problems. However, the effectiveness of collaborative governance in delivering sustainable solutions is variable, ranging from highly effective to potentially insignificant, depending on the specific circumstances [14]. This suggests that CEG’s success may be context-dependent and underscores the importance of skillful management of collaborative processes. Furthermore, there is a recognized need for skillful bridging between different actors and initiatives within the CEG framework. Government entities, in particular, have a crucial role in facilitating collaborative approaches, as inferred from the perspectives of environmental professionals, scholars, government representatives, and community influencers [15]. This highlights the vital role of government institutions in creating ecosystems for collaborative engagement and effective environmental governance.

Accordingly, this study defines CEG as an integrative approach to environmental challenges that relies on the collaboration of diverse stakeholders, including the government, the private sector and the general public. It leverages collective strengths to create innovative and context-dependent solutions, with its effectiveness often enhanced by skillful facilitation, particularly from governmental institutions.

2.2. Empirical Studies of CEG

In recent decades, researchers and other observers in the field of environmental governance have seen a significant paradigm shift towards collaboration, referred to as collaborative environmental governance (CEG). CEG has garnered significant attention in

academic research. Scholars have emphasized the importance of considering both “collaborative governance” and “governing collaboration” in cross-sector collaborations, as these concepts present challenges and tensions that must be addressed to sustain the governance form and maximize collaborative advantages [16,17]. The emergence of these collaborations often necessitates an understanding of the institutional contexts in which they develop. The exploration of collaborative watershed partnerships presents an interesting case, revealing how varied institutional structures and political environments can influence the collaborative process. Notably, the political contracting framework (PCF) has offered insights into the formation and effectiveness of such partnerships in the United Kingdom [18].

The main component of CEG is the integration of diverse stakeholders from the general public and the private and non-government sectors. Some studies emphasize the potential of these collaborations to achieve shared environmental goals that may otherwise be unattainable for individual stakeholders [12,19]. Endeavors to overcome barriers, foster cross-sector collaboration, and bridge disparate actors have been considered instrumental in enhancing the outcomes of environmental governance initiatives [15,20].

Empirical studies have attempted to investigate the practical aspects of CEG. Taylor and de Loe [21] conducted qualitative research in South Australia that highlighted the complexities of water allocation planning. Their findings indicated that this planning involved multiple dimensions of local knowledge and emphasized the varying nature and quality of different knowledge types. Case studies have demonstrated the potential of collaborative environmental governance in achieving agreed policy outcomes for contested common-pool resources [22]. Environmental co-governance, in conjunction with established environmental regulations, was found to reduce the likelihood of enterprise migration, limit pollution transfer, and improve the efficiency of environmental governance [23].

Moreover, Avoyan [24] offered a new perspective on environmental management, employing a qualitative comparative analysis (QCA) to examine the conditions that give rise to innovative solutions. CEG, despite its potential advantages, is not immune to deadlock. A study by Cristofoli et al. [25] investigated the dynamics of stalemates within CEG. Their findings suggested that overcoming impasses necessitates a blend of reinvigorating stakeholder engagement, addressing concerns about becoming marginalized, and involving leaders able to orchestrate the process of reaching a consensus. While the literature has offered comprehensive analyses of CEG, some aspects warrant further exploration. For instance, the initial formation of collaborative relationships and the interplay of various knowledge forms within these processes have remained largely understudied [26,27]. Similarly, the dynamics of representation within these collaborations require further inquiry [28].

Meanwhile, scholars have emphasized the importance of considering complex social-ecological interdependencies and developing empirically informed theories to enhance environmental governance practices [29]. For instance, Baird et al. [5] highlighted the importance of understanding the causal mechanisms and factors to be considered when assembling coalitions for natural resource management and governance. The integration of various knowledge types and effective decision-making processes has also been found to contribute to more legitimate and effective environmental governance [3]. The role of public-private partnerships in advocating for environmental management and the impact of participatory practices on policy implementation and legitimacy was also explored [30,31].

Furthermore, insights from discursive politics and an understanding of technological framing were emphasized as enhancing cooperative environmental governance [32]. The selection of appropriate governance modes based on environmental problem characteristics was identified as being crucial for policy effectiveness [33]. Additionally, recent scholarship has ventured into the sphere of CEG within authoritarian political structures. A case study from Vietnam revealed the dominance of technocratic solutions over deliberation and the integration of local knowledge, offering fresh insights into the effects of political structure on CEG [34].

While numerous studies have explored the concept, challenges, and benefits of CEG, less focus has been directed toward factors that influence the implementation and effectiveness of CEG in these unique contexts [35]. They emphasize the need for inclusive participation, bridging actors and sectors, considering various forms of knowledge, and developing context-specific approaches to enhance environmental governance outcomes. While these studies provide valuable insights and contribute to our understanding of CEG, there is a need for quantitative research that explores the unique contextual factors, barriers, facilitators, and dynamics specific to developing countries. Accordingly, a notable gap in the current research is the lack of quantitative studies that specifically investigate the factors associated with CEG, particularly within the context of developing countries. This gap is particularly significant given that factors such as governance structures, institutional capacities, stakeholder dynamics, cultural norms, and resource constraints can have a substantial association with CEG initiatives. Therefore, there is a need for research that delves into these contextual factors and their influence on CEG within developing countries.

Addressing the research gap described above, this study aims to explore the specific factors associated with the degree of CEG in developing countries, using Thailand as a case study. By providing a comprehensive analysis of these factors, the study aims to expand our understanding of the challenges and opportunities associated with implementing effective CEG in the developing country context. Furthermore, we expect it to contribute to the existing literature on CEG by offering valuable insights into the unique contextual nuances that characterize developing countries. In so doing, we aim to guide the design and implementation of future CEG initiatives within similar contexts.

3. Hypothesis Development

The theoretical foundation of our study is based on the understanding of the different socio-economic factors that might impact the degree of CEG in Thailand. We present hypotheses to substantiate the interplay between each independent variable and CEG, as depicted in the conceptual model in Figure 1.

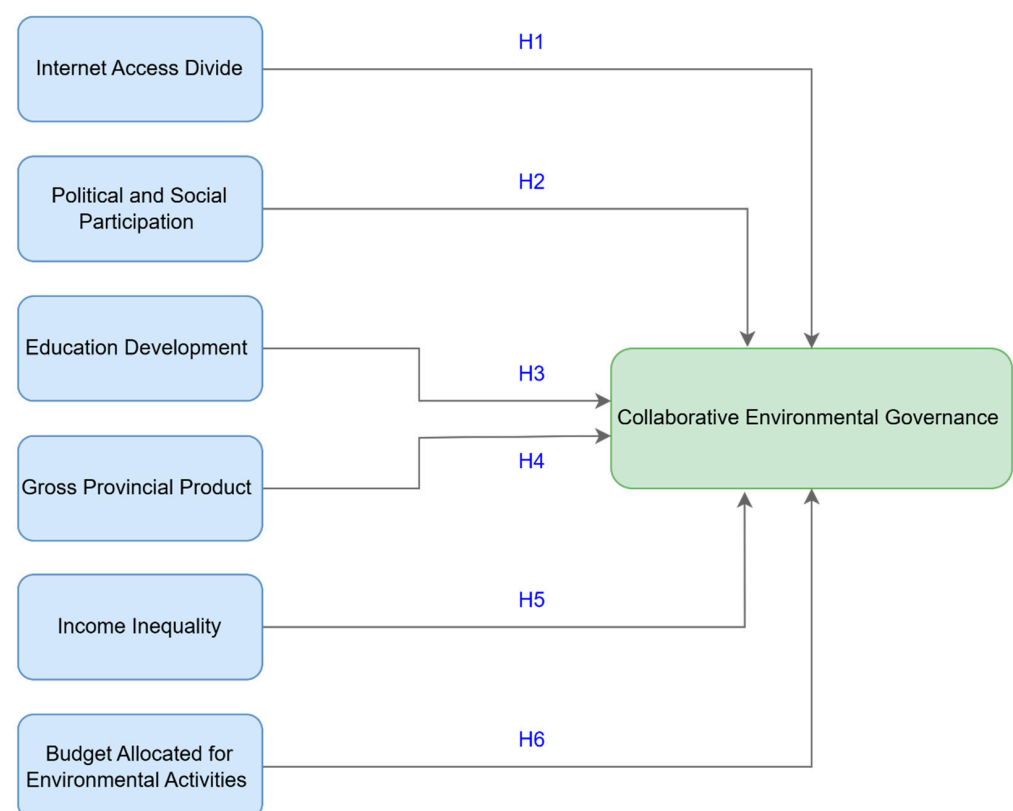


Figure 1. Conceptual model to analyze the socio-economic determinants of CEG in Thailand.

Internet Access Divide (IAD): Building on scholarly work that highlights the connection between internet usage and perceptions of environmental risks [36], we emphasize the power of the Internet as a tool for fostering increased and dynamic social interactions and collaborations [37]. These interactions may include discussion of environmental issues, shared initiatives to promote sustainable practices, and digital platforms for environmentally conscious communities. Accordingly, provinces with a wider IAD are hypothesized to display reduced levels of CEG. We expect provinces with a higher IAD, indicating less internet usage, to have lower levels of CEG, as access to information and communication technologies facilitates environmental awareness and collaborative governance. Therefore, we hypothesize the following:

H1: *The Internet Access Divide negatively influences the degree of CEG.*

Political and Social Participation Index (PSI): Drawing on the research conducted by Wang et al. [38] and Newman et al. [39], we recognize that public participation in environmental governance can address local environmental issues more effectively than relying solely on top-down approaches. However, this requires a conducive political space, sufficient funding, and a culture encouraging public participation. Their research highlights the constraints on ‘political opportunity structures’ and the ensuing limits to ‘collaborative governance’, underscoring the essential role of cooperative norms in promoting political participation [40]. Therefore, we expect that a higher degree of political and social participation in a province would engender a corresponding increase in the level of CEG. Active participation in the political and social sphere often reflects a willingness to engage in matters of environmental governance, enhancing its collaborative nature. Consequently, we posit the following:

H2: *The Political and Social Participation Index positively influences the degree of CEG.*

Education Development Index (EDI): Past research has consistently found that higher education can be employed to overcome environmental challenges [41], while environmental education was identified as pivotal in raising a generation that is more conscious of environmental problems [42]. For instance, Zsóka et al. [43] underscored a potent association between the depth of environmental education and students’ understanding of environmental issues. As such, we expect that an educated population is likely to demonstrate a greater understanding and awareness of environmental issues, which should in turn contribute positively to CEG. Accordingly, we hypothesize the following:

H3: *The Education Development Index positively influences the degree of CEG.*

Gross Provincial Product (GPP): The existing literature reveals a positive correlation between state income levels and environmental outputs [44] and researchers have argued that there is a strong observed correlation between income and governance [45]. We assume that wealthier provinces (with higher GPP) have more resources to invest in environmental governance, which should lead to a higher degree of CEG. Therefore, we hypothesize the following:

H4: *Gross Provincial Product positively influences the degree of CEG.*

Gini Coefficient (GINI): It is important to consider the impact of income disparity on CEG. Various studies have found that higher income inequality can detrimentally affect environmental sustainability and quality [46,47]. We assume that if the income disparity in a province is large—reflected in a higher Gini Coefficient—it could have a negative impact on CEG. Large income disparities could lead to social tension, thereby impeding collaborative efforts towards environmental governance. Hence, we hypothesize the following:

H5: *The Gini Coefficient negatively influences the degree of CEG.*

Budget Allocated for Environmental Activities (BAE): Mischen [48] posits that the success of collaborative networks relies heavily on the availability of essential social, intellectual, and financial resources, as well as the aptitude to administrate these assets via collaborative governance. Winsor [49] further implies that result enhancement is achievable through the optimization of financial contributions and non-monetary resources, which will also reinforce collaborative bonds among various project stakeholders. We expect that a larger budget for environmental activities at the provincial level would promote CEG. More resources mean that more activities can be implemented to promote collaboration in environmental governance. As such, we hypothesize the following:

H6: *Budget Allocated for Environmental Activities positively influences the degree of CEG.*

4. Methodology

4.1. Data Sources

To examine the factors affecting the CEG degree in Thailand, we used the panel dataset for Thailand for the years 2017 and 2019. Table 1 provides a summary of the variables used in the study, their categories, descriptions, and the sources from which the data were obtained. The dependent variable in this study is the degree of Collaborative Environmental Governance (CEG) in each province, which was calculated based on a method outlined by Duan et al. [1]. Detailed information about the calculation will be presented in the next section.

Table 1. Variable descriptions and sources.

Variable	Category	Description	Source
Dependent Variable Collaborative Environmental Governance Degree (CEG)		Degree of CEG in each province, calculated based on Duan et al. [1]	More details will be explained in the next section
Independent Variable Internet Access Divide (IAD)	Social	The ratio of people not using the Internet to total population in each province	National Statistical Office of Thailand
Political and Social Participation Index (PSI)	Social	Index that represents the levels of political and social participation in each province (e.g., participating in general elections and referenda, being a member of various community groups, and volunteering to participate in socially beneficial activities)	Office of the National Economics and Social Development Council
Education Development Index (EDI)	Social	Index that reflects the educational development of Thais both quantitatively and qualitatively. It covers four main indicators such as average number of academic years, attendance rate, average intelligence level (IQ), and average score of the O-Net test.	Office of the National Economics and Social Development Council
Gross Provincial Product (GPP)	Economic	Value of final production of goods and services of the province from all kinds of production and service activities produced within each province.	Office of the National Economics and Social Development Council
Gini Coefficient (GINI)	Economic	Values representing household income differences in each province	Office of the National Economics and Social Development Council
Budget Allocated for Environmental Activities (BAE)	Economic	The amount of government budget allocated for environmental activities in each province.	Ministry of Natural Resources and Environment

The study incorporates six independent variables, divided into social and economic categories. In the social category, the Internet Access Divide (IAD) is a ratio representing the proportion of non-internet users to the total population in each province, sourced from the National Statistical Office of Thailand. IAD was calculated based on the method proposed by Setthasuravich and Kato [50,51]. The Political and Social Participation Index (PSI) is a measure of political and social engagement in each province, encompassing activities such as voting in elections and membership in community groups, obtained from the Office of the National Economics and Social Development Council. The Education Development Index (EDI) reflects the development of education in Thailand both quantitatively and qualitatively, using such indicators as academic years, attendance rate, average IQ, and O-Net test scores, sourced from the same office as the PSI.

The economic category includes the Gross Provincial Product (GPP), which represents the total value of final goods and services produced in each province, obtained from the Office of the National Economics and Social Development Council. The Gini Coefficient (GINI) indicates income inequality within each province, and the budget allocated for environmental activities (BAE) signifies the amount of government budget dedicated to environmental initiatives. Those figures are sourced from the Ministry of Natural Resources and Environment.

4.2. The CEG Index System

Table 2 outlines the index system developed by Duan et al. [1] and is used to calculate the degree of CEG in each province. The system is hierarchical and divided into primary subsystems, secondary subsystems, and specific parameters, with appropriate data sources specified for each parameter. The CEG index comprises three primary subsystems: the government, the private sector, and the general public. This reflects the involvement of these sectors in environmental governance.

Within the government subsystem, the study examines regulatory aspects by determining the number of issued local laws, regulations, rules, administrative review organizations, and administrative punishment agencies. The parameter “Number of issued local laws and regulations related to environmental concerns” represents the formal legal framework governing environmental issues at higher administrative levels and encompasses municipal ordinances and those issued by provincial administrative organizations. The data for this parameter are sourced from the Department of Local Administration. The “Number of issued local rules related to environmental concerns” quantifies the ordinances issued by sub-district administrative organizations, which address environmental concerns at a more localized level. The data are also obtained from the Department of Local Administration. The number of administrative review organizations related to environmental issues, such as the Provincial Damrongdhama Center, in each province is obtained from the Ministry of Interior. Furthermore, the number of administrative punishment agencies responsible for enforcing penalties for violations of environmental laws, including the Provincial Prosecutor and Provincial Court, is sourced from the Ministry of Justice. The administrator category within the government subsystem reflects the role of personnel in various administrative departments. The “Number of administrative departments specifically dedicated to environmental protection” within provincial governments is determined from data provided by the Ministry of Natural Resources and Environment. The “Number of personnel at environmental monitoring agencies” measures the workforce involved in agencies that monitor environmental conditions and compliance based on data from the Ministry of Natural Resources and Environment. Similarly, the “Number of personnel at environmental monitoring stations” is a count of the staff at stations responsible for on-the-ground environmental monitoring activities from data procured from the Ministry of Industry. Additionally, the “Number of education centers” parameter counts the centers focusing on environmental education using data from the Equitable Education Fund.

In the private subsystem, the study considers the role of corporations participating in environmental governance. The parameter “Number of green corporations approved by

the Ministry of Industry” measures the involvement of the private sector in environmental governance in terms of the number of corporations recognized for their environmentally friendly practices. This data is sourced from the Ministry of Industry.

Table 2. CEG index system.

System	Primary Subsystem	Secondary Subsystem	Parameters	Sources
CEG	Government	Regulation	Number of issued local laws and regulations related to environmental concerns (e.g., municipal ordinances, provincial administrative organization ordinances).	Department of Local Administration
			Number of issued local rules related to environmental concerns (e.g., sub-district administrative organization ordinances)	Department of Local Administration
			Number of administrative review organizations related to environmental issues in each province (e.g., Provincial Damrongdhama Center).	Ministry of Interior
			Number of administrative punishment agencies related to environmental issues in each province (e.g., Provincial Prosecutor and Provincial Court).	Ministry of Justice
		Administrator	Number of administrative departments specifically dedicated to environmental protection	Ministry of Natural Resources and Environment
			Number of personnel at environmental monitoring agencies	Ministry of Natural Resources and Environment
			Number of personnel at environmental monitoring stations	Ministry of Industry
			Number of education centers	Equitable Education Fund
	Private	Participating in environmental governance	Number of green corporations approved by the Ministry of Industry	Ministry of Industry
	General Public	Environmental petition	Number of environmental petitions	Ministry of Natural Resources and Environment
			Number of personnel handling environmental petitions	Ministry of Natural Resources and Environment
		Environmental proposals	Number of national pollution control proposals	Ministry of Natural Resources and Environment
			Number of national environmental quality proposals	Ministry of Natural Resources and Environment

Finally, the general public subsystem includes indicators of public engagement in environmental governance. The “Number of environmental petitions” counts the number of petitions filed by the public to address environmental issues, reflecting public engagement and advocacy. These data are sourced from the Ministry of Natural Resources and Environment. Additionally, the “Number of personnel handling environmental petitions” measures the workforce dedicated to managing and processing these petitions and is also based on data sourced from the Ministry of Natural Resources and Environment. The

“Environmental proposals” category includes the “Number of national pollution control proposals”, which reflects proactive measures taken by the public and stakeholders to control pollution at the national level, and the “Number of national environmental quality proposals”, which measures the number of proposals aimed at improving environmental quality. The data for both parameters are sourced from the Ministry of Natural Resources and Environment.

4.3. Measuring the CEG Degree

In order to calculate the CEG degree for each province, this study used the index system (see Table 2) developed by Duan et al. [1], which could systematically assess the degree of CEG in 4 main stages. First is the process of standardizing the original data to eliminate the influence of dimension, magnitude, and positive and negative orientation. The standardized value of each indicator is calculated using Equations (1) and (2).

Positive variable:

$$y_{ij} = \frac{[x_{ij} - \min(x_{ij})]}{[\max(x_{ij}) - \min(x_{ij})]} \quad (1)$$

Negative variable:

$$y_{ij} = \frac{[\max(x_{ij}) - x_{ij}]}{[\max(x_{ij}) - \min(x_{ij})]} \quad (2)$$

The sample values denoted by x_{ij} , where i represents the year and j represents the dimension (i.e., primary subsystem) of government, private or general public. The maximum and minimum values in the sample data are denoted by $\max(x_{ij})$ and $\min(x_{ij})$, respectively. For the positive indicator, the range is calculated by subtracting the minimum value from the maximum value. The computed value for both cases ranged from 0 to 1. In the case of a positive indicator equation, higher values denote a higher degree of the particular variable. Conversely, for the negative indicator equation, higher values represent a lower degree of the specific variable.

Second, the weight of each index is determined using the entropy method. Equations (3) through (6) are used for the computation of these index weights.

The proportion of the variable j in year i can be defined as:

$$Y_{ij} = \frac{X'_{ij}}{\sum_{i=1}^m X'_{ij}} \quad (3)$$

where X'_{ij} is the standardized value of the indicator j in year i , and m is the number of years.

The calculation of information entropy for the indicator proceeds as follows:

$$e_j = \left[\frac{-1}{\ln(m)} \right] \sum_{i=1}^m Y_{ij} \ln Y_{ij} \quad (0 \leq e_j \leq 1) \quad (4)$$

where e_j is the entropy of the indicator j .

The entropy repetition is calculated using the following formula:

$$r_j = 1 - e_j \quad (5)$$

where r_i stands the entropy repetition of the indicator j .

The weight of the indicator can be expressed as:

$$w_j = \frac{r_j}{\sum_{j=1}^n r_j} \quad (6)$$

Third, we can calculate each year's level of CEG (i.e., single indicator and comprehensive level).

We calculate a single indicator as follows:

$$C_{ij} = w_j \times X'_{ij} \quad (7)$$

Calculating a comprehensive level for each subsystem (i.e., government ($Cgov_i$), private ($Cpri_i$), and general public ($Cpub_i$)) in CEG in year i can be expressed as:

$$C_i = \sum_{j=1}^n C_{ij} \quad (8)$$

Fourth, in order to calculate CEG degree after attaining the comprehensive level of each subsystem from Equation (8), we still employed the process developed by Duan et al. [1] and Song et al. [52] to evaluate. They provide a calculation formula as follows:

$$B_i = \frac{3[Cgov_i \times Cpri_i \times Cpub_i]}{[Cgov_i \times Cpri_i \times Cpub_i]^2} \quad (9)$$

However, the Formula (9) may result in a higher coordination degree even if the three subsystems are at a lower level, which we consider a drawback. To address this issue, we modify the formula, resulting in the following equations:

$$CL_i = \frac{1}{2} (Cgov_i) + \frac{1}{4} (Cpri_i) + \frac{1}{4} (Cpub_i) \quad (10)$$

$$D_i = B_i \times CL_i \quad (11)$$

In these equations, the collaboration between the three subsystems in year i is represented by B_i , the comprehensive level of the three subsystems in year i is denoted as CL_i , and the collaboration degree between the three systems in year i is indicated by D_i . The comprehensive level of environmental governance of government, private, and the general public in year i are represented by $Cgov_i$, $Cpri_i$, and $Cpub_i$, respectively. The weights that appear in Equation (10) determine the relative importance of each subsystem in the calculation. In this study, considering the dominant role of the government in CEG, the weights are assumed to be $\frac{1}{2}$ for government, $\frac{1}{4}$ for the private sector, and $\frac{1}{4}$ for the general public.

4.4. Model Specification

Our study uses panel data for 77 provinces in Thailand from the years 2017 and 2019. Our aim is to investigate determinants of the CEG. We used three different panel data estimation techniques: Pooled Ordinary Least Squares (OLS), Random Effects (RE), and Fixed Effects models (FE). The choice of the best fitting model was guided by statistical tests such as the Hausman specification test. Panel data analysis helps to control unobservable constants over time, thus strengthening the validity of any inferences drawn regarding relationships of interest [53]. All variables were natural logarithm-transformed ($\ln$) to normalize the distribution and interpret the coefficients as elasticities.

4.4.1. Pooled Ordinary Least Squares (OLS)

OLS is a model that assumes that the coefficients of the independent variables are the same for all individuals or entities in the panel. This model is appropriate when there is no individual-specific effect that affects the dependent variable. OLS model can be represented as follows:

$$\ln\text{CEG}_{it} = \beta_0 + \beta_1 \ln\text{IAD}_{it} + \beta_2 \ln\text{PSI}_{it} + \beta_3 \ln\text{EDI}_{it} + \beta_4 \ln\text{GPP}_{it} + \beta_5 \ln\text{GINI}_{it} + \beta_6 \ln\text{BAE}_{it} + \varepsilon_{it} \quad (12)$$

where $\ln\text{CEG}_{it}$ represents the degree of CEG in province i in year t . The term $\ln\text{IAD}_{it}$ represents the natural logarithm of the Internet Access Divide in province i in year t , reflecting the ratio of non-internet users to the total population in each province. $\ln\text{PSI}_{it}$ denotes the natural logarithm of the Political and Social Participation Index in province i in year t ; this index measures the levels of political and social engagement in each province. Similarly, $\ln\text{EDI}_{it}$ represents the natural logarithm of the Education Development Index in province i in year t . This index includes indicators such as academic years, attendance rate, average IQ, and O-Net test scores. Furthermore, $\ln\text{GPP}_{it}$ is the natural logarithm of the Gross Provincial Product in province i in year t . The term $\ln\text{GINI}_{it}$ is the natural logarithm of the Gini Coefficient in province i in year t and reflects income inequality within each province. $\ln\text{BAE}_{it}$ denotes the natural logarithm of the budget allocated for environmental activities in province i in year t . In the equations, β_0 is the intercept, and $\beta_1, \beta_2, \beta_3, \beta_4, \beta_5$, and β_6 are the coefficients for the respective variables. Finally, ε_{it} is the error term.

4.4.2. Random Effects Model (RE)

In a random effects model, province-specific effects are considered as random variables uncorrelated with the independent variables, permitting the inclusion of time-invariant variables. A critical premise of the RE model is the non-correlation between the error term and independent variables, an assumption that may not hold in every circumstance. Notably, the RE model presumes that the newly added categorical term does not correlate with the independent variables, unlike the FE model, which assumes such a correlation. According to Zhu [54], in the RE model, the newly added unit-specific terms are considered independent from the pooled cross-sectional units across all time periods, meaning that individual characteristics are assumed to exert no influence on the regressors. The RE model can be expressed as:

$$\ln\text{CEG}_{it} = \beta_0 + \beta_1 \ln\text{IAD}_{it} + \beta_2 \ln\text{PSI}_{it} + \beta_3 \ln\text{EDI}_{it} + \beta_4 \ln\text{GPP}_{it} + \beta_5 \ln\text{GINI}_{it} + \beta_6 \ln\text{BAE}_{it} + u_i + \varepsilon_{it} \quad (13)$$

where u_i is the province-specific random effect and ε_{it} is the idiosyncratic error term.

4.4.3. Fixed Effects Model (FE)

Conversely, a fixed effects model posits that the individual-specific effects (in our case, province-specific effects) are non-random and constant. This model primarily isolates the impact of time-invariant characteristics, allowing the assessment of the predictors' net effect on the dependent variable. It assumes that the error term is uncorrelated with the independent variables, a premise that is less restrictive than that in the Random Effects model. The FE model can be expressed as:

$$\ln\text{CEG}_{it} = \beta_0 + \beta_1 \ln\text{IAD}_{it} + \beta_2 \ln\text{PSI}_{it} + \beta_3 \ln\text{EDI}_{it} + \beta_4 \ln\text{GPP}_{it} + \beta_5 \ln\text{GINI}_{it} + \beta_6 \ln\text{BAE}_{it} + \alpha_i + \varepsilon_{it} \quad (14)$$

where α_i denotes the province-specific fixed effects, and ε_{it} represents the idiosyncratic error term.

To choose between employing a Fixed Effects (FE) or a Random Effects (RE) model, the Hausman test is employed. This test evaluates whether unique errors are correlated with the regressors; the null hypothesis is that they are uncorrelated. If the null hypothesis is rejected, it suggests that a Fixed Effects model is more appropriate. Conversely, if the null hypothesis is accepted, a Random Effects model is the better choice.

5. Results

Table 3 presents the descriptive statistics of the variables employed in this study. $\ln\text{CEG}$ has an average value of 0.722 with a standard deviation of 0.042, indicating vari-

ability within the data. The natural log of Internet Access Divide (*lnIAD*) has a mean of 0.875 with a standard deviation of 0.040, reflecting variation in internet usage across provinces. The average natural log of the Political and Social Participation Index (*lnPSI*) is 0.932, the natural log of the Education Development Index (*lnEDI*) averages 0.889, the natural log of the Gross Provincial Product (*lnGPP*) has a mean of 11.775, the natural log of the Gini Coefficient (*lnGINI*) averages 0.837, and the natural log of the Budget Allocated for Environmental Activities (*lnBAE*) averages 8.224. Each variable's Variance Inflation Factor (VIF) is below the common threshold of 10, indicating no severe multicollinearity [55].

Table 3. Descriptive statistics.

Variable	Obs	VIF	Mean	Std. Dev.	Min	Max
<i>lnCEG</i>	154	-	0.722	0.042	0.693	0.910
<i>lnIAD</i>	154	1.11	0.875	0.040	0.783	0.995
<i>lnPSI</i>	154	1.36	0.932	0.050	0.766	1.055
<i>lnEDI</i>	154	1.77	0.889	0.056	0.751	1.074
<i>lnGPP</i>	154	2.28	11.775	0.648	10.931	13.833
<i>lnGINI</i>	154	1.25	0.837	0.031	0.747	0.936
<i>lnBAE</i>	154	1.02	8.224	0.469	6.566	9.867

Note: Std. Dev. denotes standard deviation; VIF represents variance inflation factor.

Table 4 shows the paired *t*-test results for each variable, which indicate whether there is a statistically significant difference in the mean values of that variable between the years 2017 and 2019. For the variable *lnCEG*, the mean difference is -0.009 with a *t*-statistic of -2.199 and a *p*-value of 0.031, indicating a statistically significant decrease in the degree of CEG between 2017 and 2019. For *lnIAD*, the mean difference is 0.054 with a *t*-statistic of 28.014 and a *p*-value of 0.000, suggesting a highly significant increase in the Internet Access Divide over the same period. The variable *lnPSI* shows a mean difference of 0.006, a *t*-statistic of 1.637, and a *p*-value of 0.1058, indicating no statistically significant change in the Political and Social Participation Index from 2017 to 2019. For *lnEDI*, the mean difference is 0.004 with a *t*-statistic of 3.228 and a *p*-value of 0.002, reflecting a statistically significant increase in the Education Development Index. The *lnGPP* variable has a mean difference of -0.063 , a *t*-statistic of -8.394 , and a *p*-value of 0.000, indicating a highly significant decrease in the gross provincial product. The variable *lnGINI* shows a mean difference of 0.006, a *t*-statistic of 2.124, and a *p*-value of 0.037, suggesting a statistically significant increase in income inequality. Finally, for *lnBAE*, the mean difference is 0.145 with a *t*-statistic of 4.732 and a *p*-value of 0.000, indicating a highly significant increase in the environmental budget from 2017 to 2019.

Table 4. Paired *t*-test results.

Variable	Obs	Mean (2017)	Mean (2019)	Mean Difference	Std. Error of Diff.	t-Statistic	df	p-Value (Two-Tailed)
<i>lnCEG</i>	77	0.717	0.726	-0.009	0.004	-2.199	76.000	0.031
<i>lnIAD</i>	77	0.902	0.848	0.054	0.002	28.014	76.000	0.000
<i>lnPSI</i>	77	0.935	0.930	0.006	0.003	1.637	76.000	0.106
<i>lnEDI</i>	77	0.891	0.887	0.004	0.001	3.228	76.000	0.002
<i>lnGPP</i>	77	11.744	11.807	-0.063	0.008	-8.394	76.000	0.000
<i>lnGINI</i>	77	0.840	0.835	0.006	0.003	2.124	76.000	0.037
<i>lnBAE</i>	77	8.297	8.151	0.145	0.031	4.732	76.000	0.000

These paired *t*-tests offer a preliminary understanding of the data, providing initial insights into the significance and direction of changes in key indicators over time and highlighting areas of potential interest for further analyses. They also effectively highlight methodological disparities and provide a solid foundation for a comprehensive examination of the relationships among variables in the panel data analysis. Panel regression analysis provides a comprehensive and robust examination of relationships over time, accounting for individual heterogeneity and dynamic effects. Panel data analysis allows us to control for unobserved individual heterogeneity (e.g., characteristics of provinces) that do not change over time but could affect the dependent variable. Additionally, it can be used to explore how changes in explanatory variables over time affect the dependent variable, providing a more dynamic understanding of the relationships between variables.

Table 5 presents the results of panel data analysis using OLS, RE, and FE models, employing six key independent variables to predict the Collaborative Environmental Governance (CEG) in Thailand's provinces. Results show that for the social category variables, it was found that only Internet Access Divide (*lnIAD*) is associated with CEG degree. *lnIAD* is negatively associated with CEG degree in all models, suggesting that a higher Internet Access Divide reduces CEG degree. Specifically, the coefficient is negative across all models. It is only significant at the 10% level in the OLS and at the 5% level in the RE and FE models, indicating that a 1% increase in the Internet Access Divide leads to a decrease in the degree of CEG by approximately 0.126% to 0.154%, depending on the model. This suggests that better internet access or usage might facilitate more CEG, possibly by enabling better communication, information dissemination, or awareness of environmental issues and initiatives.

Table 5. Estimation results: comparing pooled OLS, Random Effects, and Fixed Effects models.

Variable	(1) Pooled OLS (OLS)		(2) Random Effects (RE)		(3) Fixed Effects (FE)	
	Coef. (t-Score)	Robust SE	Coef. (z-Score)	Robust SE	Coef. (z-Score)	Robust SE
<i>lnIAD</i>	−0.126 ■ (−1.96)	0.064	−0.135 * (−2.30)	0.059	−0.154 * (−2.36)	0.065
<i>lnPSI</i>	0.005 (0.09)	0.056	0.040 (0.65)	0.062	0.154 * (2.03)	0.075
<i>lnEDI</i>	0.085 (1.31)	0.065	0.089 (1.10)	0.081	−0.044 (−0.08)	0.557
<i>lnGPP</i>	0.025 *** (4.11)	0.006	0.025 ** (3.14)	0.008	−0.041 (−1.38)	0.030
<i>lnGINI</i>	−0.278 ** (−2.79)	0.010	−0.283 * (−2.44)	0.116	−0.237 (−1.17)	0.202
<i>lnBAE</i>	0.022 *** (3.76)	0.006	0.017 ** (3.27)	0.005	−0.005 (−0.68)	0.007
Intercept	0.509 *** (3.96)	0.129	0.516 *** (3.87)	0.133	1.477 * (2.58)	0.571
N	154		154		154	
Adjusted <i>R</i> ²	0.410		—		—	
Within <i>R</i> ²	—		0.049		0.115	
Between <i>R</i> ²	—		0.518		0.325	
Overall <i>R</i> ²	—		0.431		0.226	

Note: *t* statistics and *z* statistics in parentheses, robust SE denotes the robust standard error, and ■ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

The Political and Social Participation Index (*lnPSI*) showed varied results across the models. *lnPSI* is a significant positive coefficient in the FE model implying that a 1% increase in political and social participation is associated with an approximate 0.154% increase in the degree of CEG. This indicates that in contexts where time-invariant charac-

teristics are controlled, higher political and social participation is associated with a higher degree of CEG. This aligns with theories that advocate for active citizen engagement in governance processes to improve outcomes, while the Education Development Index (*lnEDI*) is not significant across all models, indicating that there is insufficient evidence of it influencing CEG.

With regard to the economic category variables, we found that all variables influenced the CEG degree only in the OLS and RE models. For example, Gross Provincial Product (*lnGPP*) is positively correlated with CEG degree. The coefficient is positive and statistically significant at the 0.1% level in the OLS model and at the 1% level in the RE model. This implies that a 1% increase in *lnGPP* is associated with a 0.025% increase in the degree of CEG, indicating a strong positive relationship between *lnGPP* and CEG. This result suggests that provinces with higher economic activity or wealth levels are likely to have a higher degree of CEG. It could be that wealthier provinces have more resources to invest in environmental governance.

In addition, we also found that the Gini Coefficient (*lnGINI*) is negatively associated with CEG degree, indicating that income inequality reduces CEG. The coefficient is negative across all models, suggesting a negative relationship with CEG. This indicates that a 1% increase in income inequality leads to a decrease in CEG by approximately 0.278% to 0.283%. A possible explanation could be that areas with greater income inequality may experience more social conflict or less public consensus, which could hinder the establishment of CEG. Moreover, the results illustrated that the Budget Allocated for Environmental Activities (*lnBAE*) is positively related to the CEG degree, suggesting that a greater budget for environmental activities enhances CEG. The coefficient is positive and significant at the 0.1% level in the OLS model and at the 1% level in the RE model, indicating a positive relationship with CEG in these models. This suggests that a 1% increase in the budget for environmental activities is associated with a 0.022% to 0.017% increase in CEG. This finding indicates that the provinces with greater government budget allocations for environmental activities have higher CEG. This is expected, as greater resources specifically devoted to environmental activities are likely to support better environmental governance.

From these results, one could conclude that the OLS and RE models perform better in terms of the statistical significance of the predictors and the R^2 values, while the FE model appears to fit the data less well. However, to choose whether RE or FE models are most appropriate, we use the Hausman test. The results show that $\chi^2(6) = 9.11$. The p -value is 0.1678, which is greater than 0.05. Thus, we fail to reject the null hypothesis. In other words, the test provides no evidence to reject the RE model in favor of the FE model. This implies that the RE model may be a more appropriate choice.

6. Robustness Test

In the initial model, we anticipated that the government's role in CEG would be more influential than those of the private sector and general public. This assumption was predicated on the government's regulatory and policy-making capacities, which often play a pivotal role in shaping environmental governance frameworks. To critically assess the robustness of our results against this assumption, we conducted a sensitivity analysis by adjusting the weights attributed to each of the three subsystems—the government, the private sector, and the general public—in the CEG index calculation. We modified the weight distribution in Equation (10) to allocate equal importance to the subsystems, assigning each a weight of 0.333. This adjustment was performed to test whether the influence of the government is disproportionately significant compared to that of the other sectors and whether a more balanced model would yield similar results.

The modified results are detailed in Table 6. Despite the altered weight distribution, the change in the model outputs was minimal. This lack of a substantial difference reinforced the stability and reliability of our original findings. The results suggest that while the government's role is indeed central, the contributions of the private sector and the general public are equally vital to the robustness of environmental governance outcomes.

This implies that our initial model is robust, even when our assumption of a dominant government influence is neutralized.

Table 6. Robustness test results: comparing Pooled OLS, Random Effects, and Fixed Effects models with different weights in CEG calculation.

Variable	(1) Pooled OLS (OLS)		(2) Random Effects (RE)		(3) Fixed Effects (FE)	
	Coef. (t-Score)	Robust SE	Coef. (z-Score)	Robust SE	Coef. (z-Score)	Robust SE
<i>lnIAD</i>	−0.111 ■ (−1.81)	0.061	−0.122 * (−2.23)	0.055	−0.160 ** (−2.65)	0.061
<i>lnPSI</i>	0.008 (0.14)	0.053	0.043 (0.74)	0.059	0.159 * (2.21)	0.072
<i>lnEDI</i>	0.074 (1.25)	0.059	0.078 (1.05)	0.074	−0.069 (−0.14)	0.486
<i>lnGPP</i>	0.027 *** (4.48)	0.006	0.027 *** (3.39)	0.008	−0.046 (−1.60)	0.029
<i>lnGINI</i>	−0.289 ** (−3.36)	0.086	−0.287 ** (−2.84)	0.101	−0.212 (−1.20)	0.177
<i>lnBAE</i>	0.017 *** (3.38)	0.005	0.013 ** (2.81)	0.005	−0.005 (−0.73)	0.007
Intercept	0.529 *** (4.38)	0.121	0.528 *** (4.19)	0.126	1.530 ** (2.90)	0.527
<i>N</i>	154		154		154	
Adjusted <i>R</i> ²	0.482		—		—	
Within <i>R</i> ²	—		0.065		0.141	
Between <i>R</i> ²	—		0.562		0.398	
Overall <i>R</i> ²	—		0.479		0.292	

Note: *t* statistics and *z* statistics in parentheses, robust SE denotes the robust standard error, and ■ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

This robustness test underscores that the dynamics of CEG are complex and inter-dependent, requiring balanced contributions from all sectors. The findings from these tests validate our theoretical framework, demonstrating that effective CEG is not solely dependent on governmental action but on a symmetrical tripartite collaboration. Due to this equilibrium, our model remains applicable across various governance structures, enhancing the generalizability of our conclusions.

7. Discussion and Conclusions

7.1. Summary of the Findings

The primary objective of this research was to explore the determining factors that influence the degree of Collaborative Environmental Governance (*lnCEG*) in Thailand. We examined a range of socio-economic indicators, including Internet Access Divide (*lnIAD*), Political and Social Participation Index (*lnPSI*), Education Development Index (*lnEDI*), Gross Provincial Product (*lnGPP*), Gini Coefficient (*lnGINI*), and the Budget Allocated for Environmental Activities (*lnBAE*). Our analysis employed panel data techniques—Pooled OLS, Random Effects, and Fixed Effects models—and covered the years from 2017 to 2019 for provinces in Thailand.

Overall, the findings indicated that the Internet Access Divide (*lnIAD*) and Gross Provincial Product (*lnGPP*) consistently showed significant correlations with CEG across all models. The negative relationship between *lnIAD* and *lnCEG* suggests that provinces with a lower proportion of internet non-users tend to have higher degrees of CEG. This could potentially be due to internet connectivity enabling increased access to environmental information and opportunities for collaborative governance. This finding aligns with existing research. For instance, Zhang et al. [36] noted that internet use significantly

affects the risk perception of environmental issues. In the same vein, Sooryamoorthy and Shrum [37] also found that internet use positively correlates with collaboration, which can enhance CEG. Therefore, inequalities in internet access can hinder the sharing of environmental information and collaborative efforts, reducing the effectiveness of CEG [56]. This relationship can be understood in the context of how the internet facilitates the flow of information, supports the mobilization of resources, and enables the participation of diverse stakeholders in environmental decision-making processes. The internet acts as a platform for knowledge exchange, advocacy, and coordination among environmental stakeholders, including government bodies, non-governmental organizations, community groups, and the general public. In regions where the digital divide is pronounced, the reduced internet access impedes these essential functions, potentially leading to less informed and less inclusive governance processes. This, in turn, can diminish the effectiveness of environmental governance by limiting stakeholder engagement and the availability of critical environmental information. The findings suggest that addressing the digital divide could be a strategic lever for enhancing CEG. Policies aimed at increasing internet accessibility and usage, especially in underserved areas, could significantly contribute to bridging the gap. Such efforts could include investing in digital infrastructure, promoting affordable internet services, and implementing digital literacy programs. Moreover, encouraging the use of digital platforms for environmental governance activities could foster more inclusive and effective participation in environmental decision-making processes.

In contrast, the positive correlation of $\ln GPP$ with $\ln CEG$ suggests that wealthier provinces tend to have a higher degree of CEG. This association is particularly significant, highlighting that provinces with higher levels of economic activity or wealth are better positioned to foster a higher degree of CEG. The underlying rationale for this relationship could be that wealthier provinces possess more financial resources, which can be allocated towards the implementation of environmental policies, projects, and initiatives. This may include investments in sustainable infrastructure, environmental monitoring, and enforcement mechanisms, as well as public awareness campaigns and educational programs on environmental governance. The findings resonate with the broader discourse in environmental economics and governance literature. Li and Chen [57] suggested that relative income is a better predictor of environmental concern than absolute income, implying that the socioeconomic status within a community influences its environmental priorities and actions. Similarly, List and McHone [44] demonstrated a positive link between income levels and environmental outcomes, supporting the notion that economic prosperity can lead to better environmental management and protection. Moreover, these results align with Meckling's [58], who suggested a strong connection between economic development and environmental governance. The idea is that economic development provides not just the resources but also the institutional and technological capacities to address environmental challenges more effectively. However, it is critical to recognize that while economic development is positively associated with CEG, this relationship is complex and multi-faceted. Economic growth should be pursued in a manner that is sustainable and does not compromise environmental integrity. Sustainable development strategies that integrate economic, environmental, and social goals are essential for fostering long-term prosperity and environmental health.

The Gini Coefficient ($\ln GINI$) and the Budget allocated for environmental activities ($\ln BAE$) showed significant effects in some models, indicating the potential role of income inequality and resource allocation in environmental governance. The negative association between the $\ln GINI$ and the degree of $\ln CEG$ is a critical finding, underlining the adverse impact of income inequality on environmental governance. Income inequality can lead to divergent interests and priorities among different social groups, making it challenging to achieve unified action towards environmental sustainability. In areas with significant income disparity, those with fewer resources may have immediate survival priorities over long-term environmental concerns, whereas wealthier segments might prioritize environmental initiatives that align with their interests. This division can stifle collaborative efforts,

as achieving common ground becomes increasingly difficult. This finding aligns with the broader literature, such as the work of Masud et al. [46], who identified a link between income inequality and environmental sustainability. Similarly, Zhou and Li [59] found that higher income inequality negatively impacts environmental quality, further affirming the negative effects of inequality on environmental governance outcomes. Conversely, the positive relationship between $\ln BAE$ and $\ln CEG$ emphasizes the important role of financial resources in environmental governance. This suggests that provinces with higher government budget allocations for environmental initiatives tend to exhibit stronger CEG. The availability of funds is crucial for implementing environmental policies, projects, and educational programs, which collectively foster a conducive environment for collaborative governance. This finding is in line with Mischen's [48] work about the importance of managing capital resources in successful collaborative networks which complements the observed positive impact of environmental budget allocation on CEG. These insights emphasize the necessity for policies that not only aim to reduce income inequality but also ensure adequate financial investments in environmental governance. Addressing income disparity could mitigate social conflicts and foster public consensus, facilitating more effective collaborative governance. Simultaneously, increasing budget allocations for environmental activities can empower provinces to implement comprehensive governance strategies. Therefore, a balanced approach that tackles both socio-economic equity and resource sufficiency could significantly enhance the prospects for collaborative environmental governance, ultimately leading to more sustainable environmental outcomes. Interestingly, the influence of the education index ($\ln EDI$) and the political and social participation index ($\ln PSI$) on $\ln CEG$ was not as pronounced as we expected. This indicates that while education and social participation play roles in societal phenomena, their direct influence on environmental governance in the Thai context might not be particularly strong.

We performed a robustness test in which we changed the weights in the CEG index calculation to be equal for the government, the private sector, and the general public; the output revealed that our main findings were robust. Despite minor variations in the magnitude of the coefficients and significance levels, the overall patterns were consistent, underscoring the stability and reliability of our findings under different weighting assumptions. This suggests that effective environmental governance relies on balanced contributions from all sectors, challenging the traditional emphasis on governmental primacy in policy frameworks. The results confirm that our conclusions are robust across different environmental governance contexts, reinforcing the theoretical implication that inclusive stakeholder participation enhances governance effectiveness. This finding further supports the broader applicability of our model and offers valuable insights for policymakers aiming to foster more effective environmental governance initiatives.

7.2. Policy Implications

Based on this study's findings, several policy recommendations can be proposed to enhance CEG in Thailand and other developing countries.

1. The government should make more effort to increase internet access, particularly in underserved areas or remote areas. This can be accomplished by improving digital infrastructure and promoting affordable internet services. Moreover, digital skills programs should be implemented to ensure the effective use of internet resources, facilitating a higher degree of public engagement in environmental governance.
2. The government may consider public policies for promoting sustainable economic growth. It can contribute to environmental governance. Such policies could involve encouraging green industries and sustainable practices among businesses, which can simultaneously stimulate economic growth and contribute to environmental sustainability. Therefore, it is essential to ensure that provinces have adequate resources to address environmental challenges. However, our results suggest that economic prosperity alone is not sufficient; targeted environmental investments are also crucial.

Our results also underline the importance of governments actively directing funds towards environmental initiatives.

3. The government should recognize the seriousness of income inequality issues, especially between Bangkok and other provinces, between urban and rural districts, and within rural areas as well. More income inequality negatively impacts CEG. To address this, policies could be formulated to reduce income disparity.

In conclusion, our research contributes to the understanding of the factors influencing CEG in Thailand. The significant variables—Internet Access Divide, Gross Provincial Product, Gini coefficient, and Budget allocated for environmental activities—highlight the crucial role of economic conditions, internet accessibility, and resource allocation in shaping CEG. As societies worldwide face the increasing urgency of environmental issues, our findings underscore the importance of bridging the digital divide, fostering economic growth, reducing income inequality, and allocating adequate resources for environmental activities to foster effective environmental governance.

7.3. Limitations and Future Research

While this study provides insightful findings into the determinants of CEG in Thailand, it has several limitations. First, unobserved heterogeneity, for instance, individual characteristics, community dynamics, or external events, and potential endogeneity issues, for example, the effectiveness of public participation may be influenced by pre-existing environmental conditions or governance structures, may limit the accuracy of our findings. Second, the scope of the study, covering only two years (2017 and 2019) and focusing solely on Thailand, restricts the generalizability of the results to other temporal and geographic contexts. The indicators selected for this study, while comprehensive, may not capture every relevant factor influencing CEG. Furthermore, the observed correlations do not necessarily imply causation, necessitating further research to establish causal links. Additionally, certain data are unavailable in Thailand, which could potentially influence the calculation of CEG. Owing to data constraints, certain socio-economic factors, cultural nuances, or local institutional dynamics might not have been captured, thereby potentially affecting the comprehensiveness and accuracy of our CEG assessment.

Future research should focus on expanding the scope of our study to include additional socio-economic variables, thereby providing a more comprehensive understanding of the determinants of Collaborative Environmental Governance in Thailand. Further studies could also extend the temporal and geographical coverage to enhance the generalizability of the findings. Innovative experimental designs could be used to address potential endogeneity issues and establish causal links. Lastly, the incorporation of data that were unavailable during this study could potentially offer further insights into the dynamics of environmental governance.

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